

# A Case-Control Study of Diet and Benign Proliferative Epithelial Disorders of the Breast<sup>1</sup>

Thomas E. Rohan,<sup>2</sup> Martin G. Cook, John D. Potter, and Anthony J. McMichael

National Cancer Institute of Canada Epidemiology Unit, McMurrich Building, 3rd Floor, University of Toronto, Toronto, Ontario, M5S 1A8, Canada [T. E. R.]; Department of Histopathology, St. George's Hospital, Blackshaw Road, London SW17, United Kingdom [M. G. C.]; Division of Epidemiology, School of Public Health, University of Minnesota, Minneapolis, Minnesota 55455 [J. D. P.]; and Department of Community Medicine, University of Adelaide, Adelaide, South Australia 5000 [A. J. M.]

## ABSTRACT

Benign proliferative epithelial disorders (BPED) of the breast are conditions which, although not proven precursors of breast cancer, are strongly associated with increased risk of this disease. In a case-control study conducted in Adelaide, South Australia, we investigated the association between dietary factors and risk of BPED. The study involved 383 cases of biopsy-confirmed BPED, 192 controls whose biopsy did not show epithelial proliferation, and 383 unbiopsied community controls matched to cases on age and area of residence. When cases were compared with community controls, there was little variation in the risk of BPED across levels of daily intake of energy, protein, and fat, but there was some suggestion of inverse associations with daily intake of retinol,  $\beta$ -carotene, and fiber; in contrast, with biopsy controls as the comparison group, risk of BPED increased with increasing energy and fat intake but varied little with retinol,  $\beta$ -carotene, and fiber intake. The results of this study provide additional evidence against roles for dietary energy and fat intake in the etiology of breast cancer. However, further studies of diet and BPED should be conducted in populations surveyed cross-sectionally for breast changes, in an attempt to limit or avoid the selection and misclassification biases to which studies of putative precursor lesions are susceptible.

## INTRODUCTION

Although dietary factors are thought to have a role in the etiology of breast cancer, the results of studies involving direct measurement of the dietary intake of individuals have mostly not provided strong support for this notion (1). One possible reason for this is that these investigations, predominantly case-control studies, have focussed on dietary intake in the recent past, whereas the critical exposure period for dietary factors may be at an earlier stage in the carcinogenic process. Further, the presence of breast cancer may cause a change in a woman's actual or perceived dietary behavior.

In the case-control study reported here, we investigated the possibility that dietary factors influence the risk of BPED<sup>3</sup> of the breast, conditions which, although not proven precursors of breast cancer, are strongly associated with increased risk of this disease (2) and are thought to have premalignant potential (3). Specific hypotheses were that risk of BPED would increase with increasing intake of energy, protein, and fat (the intakes of these three items are highly correlated) and would decrease with increasing intake of vitamin A (retinol and  $\beta$ -carotene). Specific hypotheses were not formulated for the carbohydrates, but given their major contribution to total energy intake, their association with risk of BPED was examined also.

Received 7/17/89; revised 2/2/90.

The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked *advertisement* in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

<sup>1</sup> This work was supported by grants from the Section 16 Committee of the South Australian Health Commission and from the Education or Research Appeal Fund of the Royal College of Pathologists of Australia.

<sup>2</sup> To whom requests for reprints should be addressed, at NCIC Epidemiology Unit, McMurrich Building, 3rd Floor, University of Toronto, Ontario M5S 1A8, Canada.

<sup>3</sup> The abbreviations used are: BPED, benign proliferative epithelial disorders; BBD, benign breast disease; RR, relative risk; CI, confidence interval.

## METHODS

The study methods have been described in detail elsewhere (4).

**Cases and Controls.** Women were eligible to be included in the study if they were between 18 and 75 years of age, if they resided within the metropolitan area of Adelaide, South Australia, if they were registered on the electoral roll [97.2% of persons eligible to vote are registered (5)], and if they did not have a past history of breast cancer or biopsy-confirmed BBD.

Women were eligible for selection as cases if biopsy material upon which their first histologically confirmed diagnosis of BBD was based showed evidence of epithelial proliferation (see below). Of the 506 eligible cases identified, 383 (75.7%) were enrolled, 39 (7.7%) could not be approached for interview because of refusal by the attending surgeon, 66 (13.0%) refused to be interviewed, and 18 (3.6%) were untraceable.

Two control groups were selected. One group consisted of women whose first breast biopsy, submitted to the same laboratory as that for the cases, did not show evidence of epithelial proliferation. Of the 259 women eligible for inclusion in this control group (the "biopsy control group"), 192 (74.1%) were enrolled, 17 (6.6%) could not be approached because of surgeon refusal, 39 (15.1%) refused to be interviewed, and 11 (4.3%) were untraceable. The other control group was formed by matching individually to each case one subject selected at random from the electoral roll from those women similar in age to the age of the case at diagnosis (88% within  $\pm 1$  year; the remainder within  $\pm 2-3$  years) and living in the same suburb or a suburb of similar socioeconomic profile to that of the case (6). In order to enroll the 383 community controls included in the study, a total of 582 subjects was approached (which represents a response rate of 65.8%). Reasons for nonparticipation were recorded during the latter stage of recruitment of community controls: of the 51 who did not participate, 30 refused and 21 were untraceable.

Of the 383 cases, 321 (83.8%) presented with a breast lump, 43 (11.2%) were recruited following a reduction mammoplasty, and the remaining 19 (5.0%) presented with various combinations of pain, tenderness, and nipple discharge. For the 192 biopsy controls, the corresponding figures were 121 (63.0%), 64 (33.3%), and 7 (3.6%).

**Histopathological Classification of Biopsies.** Histopathological assessment was performed on biopsy material submitted to the major private pathology laboratory in Adelaide (which receives 40% of all specimens from breast biopsies performed in Adelaide) between January 1983 and October 1985. All breast specimens were examined by one pathologist (M. G. C.) who had no knowledge of the exposure histories of the subjects. Benign nontumorous breast disease was categorized according to a system based on the methods of Bonser *et al.* (7), Page *et al.* (8), and Azzopardi (9). The group with epithelial proliferation included those with epitheliosis, radial scar, sclerosing adenosis, blunt duct adenosis, intraduct papilloma, fibroadenoma, fibroadenomatoid hyperplasia, and apocrine metaplasia. These lesions were classified further by degree of cytological atypia using a classification scheme which has been described elsewhere (4). Subjects found to have nonproliferative forms of BBD or uncomplicated cystic disease (*i.e.*, duct ectasia, columnar cell metaplasia, apocrine cysts, and atrophic cysts) formed the biopsy control group.

**Data Collection.** Study subjects were interviewed in their homes by trained interviewers who administered questionnaires in a standardized manner. Matched sets of cases and community controls were interviewed by the same interviewer. Cases and biopsy controls were interviewed as soon as possible after diagnosis. For cases, the average

interval between diagnosis and interview was 2.8 months, and for biopsy controls the average interval was 2.9 months. Each community control was interviewed as soon as possible after her corresponding case had been enrolled in the study, and the average interval between interview of cases and community controls was 1.3 months.

Sociodemographic and medical information was collected from the study subjects by use of an interviewer-administered structured questionnaire. The questionnaire sought basic biographic information, personal medical history, family history of cancer, gynecological history, reproductive history, and history of hormone use.

Information on usual dietary intake was collected from the study participants by means of a self-administered quantitative food frequency questionnaire. The questionnaire, which was designed to ascertain total daily intake of energy and of a number of nutrients, has been described in detail elsewhere (10, 11). Briefly, the questionnaire allowed recording of the usual frequency of consumption of 179 specified dietary items, provided space for participants to indicate whether their usual serving size for an item differed from that of a specified standard serving size (11, 12), and also provided space for them to indicate consumption of other items not listed in the main body of the questionnaire. Additional data recorded and used in the derivation of estimates of nutrient and energy intake (see below) included information on cooking practices; use of sugar, salt, and fat; and bread type. The food frequency questionnaire has been shown to provide estimates of nutrient and energy intake which are repeatable (13) and which are similar to those derived from studies in Australia in which dietary records have been used (10). A similar food frequency questionnaire has been shown to have repeatability and criterion validity (14).

Study subjects were instructed to record their usual dietary patterns. Thus, cases and biopsy controls were instructed to disregard any dietary changes that had occurred subsequent to diagnosis of their BBD, while community controls who had changed their dietary patterns subsequent to the date of diagnosis of their corresponding case were instructed to disregard such changes.

**Derivation of Daily Nutrient and Energy Intake.** Data on the frequency of consumption of food items were converted to daily nutrient and energy intake by a computerized dietary analysis system (10, 15). Food frequency scores for individual food items [obtained by multiplying frequency of consumption by respondent-specified usual serving size (expressed as a fraction of standard portion size)] were multiplied by the nutrient or energy content of the standard portion, after adjustments were made for the qualitative information on cooking practices mentioned above. The revised version of the McCance and Widdowson food composition tables (16), adjusted and supplemented to suit the Australian diet, was used to provide the nutrient and energy content values. Daily intake of various nutrients and of energy were then calculated by summing across all food items.

**Data Analysis.** For comparisons of cases and community controls, conditional logistic regression (17) was used to derive maximum likelihood estimates of RR and associated CI and to assess the statistical significance of the trend in risk of BPED across exposure levels. For comparisons of cases and biopsy controls, unconditional logistic regression (17) was used, with adjustment for age. For these analyses, the estimates of absolute daily nutrient and energy intake were categorized by quintiles derived from their distributions in the community control population. Initially, crude (for cases *versus* community controls) or age-adjusted (for cases *versus* biopsy controls) estimates of effect were derived for the association of each dietary factor with risk of BPED. Thereafter, estimates of effect adjusted for potential confounding were derived from multiple logistic regression models. The following variables, which had strong univariate associations with risk of BPED (in either case-community control or case-biopsy control comparisons), were included in these models as potential confounders: menopausal status, income, Quetelet's index (an index of obesity), frequency of breast self-examination, and number of medical consultations/year. The univariate and multivariate estimates of effect were generally similar, and only the latter are presented here. The sequence of analyses described above was repeated after adjusting nutrient intake for caloric intake (18). Tests for trend (on 1 d.f.) were performed by fitting the categorized indicator variables (rather than separate coefficients for

each level of the variable) into logistic regression models.

Analyses were conducted overall and in several strata of the total study population. The results of analyses performed after exclusion of peri- and postmenopausal women, after exclusion of women with a history of breast lump which had not been biopsied, and after exclusion of women whose biopsy showed fibroadenoma but no epithelial proliferation in the surrounding tissue, differed little from those based on the total study population and are not presented here. In a final set of stratum-specific analyses, the results of which are presented, risk of BPED was examined in two strata defined by degree of cytological atypia [absent or mild atypia (271 cases) and moderate or severe atypia (112 cases)]; for these analyses, intake was divided by tertiles rather than quintiles to reduce the statistical variability introduced by subdividing the data.

## RESULTS

**Characteristics of the Study Population.** Characteristics of the study population have been tabulated elsewhere (4). In brief, cases and controls were similar in terms of mean age, but cases had a lower mean Quetelet's index than either control group. A higher proportion of cases than community or biopsy controls had ever practiced breast self-examination monthly (or more frequently), while cases occupied an intermediate position between community controls and biopsy controls for income (more community controls than biopsy controls had a relatively high income), menopausal status (more community controls than biopsy controls were premenopausal), and frequency of medical consultations (biopsy controls had a higher mean number of medical consultations/year than community controls).

**Energy, Protein, and Fat.** In Table 1, cases and controls are compared with respect to daily intake of energy, protein, and fat (total fat and its constituents). With community controls as the comparison group, there were statistically significant trends of decreasing risk of BPED with increasing intake of energy, protein, and cholesterol. Although there was little variation in risk with total fat intake, point estimates of effect at relatively high levels of saturated, monounsaturated, and polyunsaturated fat were generally less than unity, but the trends in risk were not statistically significant.

When cases were compared with biopsy controls, risk of BPED increased with increasing intake of energy, total fat, and the fatty acid subgroups; for energy, total fat, and polyunsaturated fat, the trends in risk were statistically significant. Variation in risk with protein and cholesterol was relatively insubstantial.

The analyses described above were repeated with additional control for energy intake. Table 2 shows that much of the variation in risk with protein and fat intake was attributable to the strong correlation between intake of these items and energy intake. Indeed, the variation in risk observed in Table 1 was generally less pronounced after adjustment for energy intake, and most of the trends in risk were statistically nonsignificant. The one exception to this was the persistence of the statistically significant inverse association between cholesterol intake and risk of BPED when cases were compared with community controls; this was due essentially to the statistically significant reduction in risk of BPED at the uppermost level of cholesterol intake. Additionally, there was some suggestion of an inverse association between cholesterol intake and risk of BPED when cases were compared with biopsy controls.

**Retinol and  $\beta$ -Carotene.** Results for retinol and  $\beta$ -carotene are presented in Table 3. Risk of BPED decreased with increasing intake of retinol and  $\beta$ -carotene (statistically significant trends) but only when cases were compared with community controls.

Table 1 Adjusted relative risks<sup>a</sup> and 95% confidence intervals for the associations between daily energy, protein, and fat intake and benign proliferative epithelial disorders of the breast

| Dietary item           | Control group | RR (95% CI) for level: |               |               |               |               | P value for linear trend (1 d.f.) |
|------------------------|---------------|------------------------|---------------|---------------|---------------|---------------|-----------------------------------|
|                        |               | 1 (Low) <sup>b</sup>   | 2             | 3             | 4             | 5 (High)      |                                   |
| Energy                 | Community     | 1.0                    | 0.8 (0.5–1.3) | 0.8 (0.4–1.3) | 0.7 (0.4–1.2) | 0.5 (0.3–0.9) | 0.021                             |
|                        | Biopsy        | 1.0                    | 1.0 (0.6–1.7) | 1.3 (0.7–2.3) | 1.5 (0.8–2.5) | 1.6 (0.9–3.0) | 0.038                             |
| Protein                | Community     | 1.0                    | 0.9 (0.5–1.5) | 0.7 (0.4–1.2) | 0.5 (0.3–0.9) | 0.5 (0.3–0.9) | 0.003                             |
|                        | Biopsy        | 1.0                    | 0.8 (0.4–1.3) | 0.9 (0.5–1.5) | 1.1 (0.6–1.9) | 1.3 (0.7–2.3) | 0.294                             |
| Total fat              | Community     | 1.0                    | 1.8 (1.0–3.2) | 1.0 (0.6–1.8) | 1.0 (0.6–1.8) | 0.9 (0.5–1.5) | 0.142                             |
|                        | Biopsy        | 1.0                    | 1.3 (0.8–2.2) | 1.4 (0.8–2.6) | 1.5 (0.9–2.8) | 2.1 (1.2–3.8) | 0.012                             |
| Saturated fat          | Community     | 1.0                    | 1.2 (0.7–2.0) | 1.1 (0.6–1.8) | 0.9 (0.5–1.5) | 0.7 (0.4–1.2) | 0.117                             |
|                        | Biopsy        | 1.0                    | 1.0 (0.6–1.8) | 1.5 (0.8–2.7) | 0.9 (0.5–1.7) | 2.0 (1.1–3.8) | 0.065                             |
| Monounsaturated fat    | Community     | 1.0                    | 1.3 (0.8–2.3) | 0.9 (0.5–1.5) | 0.9 (0.5–1.5) | 0.7 (0.4–1.2) | 0.076                             |
|                        | Biopsy        | 1.0                    | 1.0 (0.6–1.7) | 1.0 (0.6–1.8) | 1.1 (0.6–1.9) | 1.7 (0.9–3.1) | 0.129                             |
| Polyunsaturated fat    | Community     | 1.0                    | 0.5 (0.3–0.8) | 0.8 (0.5–1.3) | 0.5 (0.3–0.8) | 0.8 (0.5–1.3) | 0.391                             |
|                        | Biopsy        | 1.0                    | 0.6 (0.3–1.0) | 1.0 (0.6–1.6) | 1.2 (0.6–2.4) | 2.2 (1.2–3.9) | 0.001                             |
| P/S ratio <sup>c</sup> | Community     | 1.0                    | 1.2 (0.7–2.1) | 0.7 (0.4–1.3) | 1.1 (0.6–1.9) | 1.1 (0.7–1.8) | 0.943                             |
|                        | Biopsy        | 1.0                    | 1.3 (0.7–2.2) | 1.0 (0.5–1.9) | 1.0 (0.6–1.8) | 1.4 (0.8–2.4) | 0.480                             |
| Cholesterol            | Community     | 1.0                    | 1.0 (0.6–1.7) | 0.8 (0.5–1.4) | 0.8 (0.4–1.3) | 0.4 (0.2–0.7) | 0.002                             |
|                        | Biopsy        | 1.0                    | 1.1 (0.6–1.9) | 1.3 (0.7–2.4) | 0.9 (0.5–1.5) | 1.5 (0.8–2.9) | 0.527                             |

<sup>a</sup> Adjusted for menopausal status, income, Quetelet's index, frequency of breast self-examination, and number of medical consultations per year.<sup>b</sup> Reference category. Quintiles for categorization of each item were as follows: energy (kJ/day; 1 Kcal = 4.2 kJ), 6136.4, 7506.5, 8654.8, 10516.1; protein (g/day), 60.9, 76.2, 90.0, 108.8; total fat (g/day), 56.3, 75.9, 90.8, 113.7; saturated fat (g/day), 21.2, 28.6, 36.1, 45.4; monounsaturated fat (g/day), 18.1, 24.6, 30.0, 38.8; polyunsaturated fat (g/day), 9.2, 13.2, 17.8, 22.3; P/S (polyunsaturated to saturated fat) ratio, 0.30, 0.44, 0.55, 0.72; cholesterol (mg/day), 179.8, 249.8, 318.7, 428.4.<sup>c</sup> Ratio of polyunsaturated to saturated fat.Table 2 Relative risks and 95% confidence intervals for the associations between energy-adjusted daily protein and fat intake<sup>a</sup> and risk of benign proliferative epithelial disorders of the breast

| Dietary item        | Control group | RR (95% CI) for level: |               |               |               |               | P value for linear trend (1 d.f.) |
|---------------------|---------------|------------------------|---------------|---------------|---------------|---------------|-----------------------------------|
|                     |               | 1 (Low) <sup>b</sup>   | 2             | 3             | 4             | 5 (High)      |                                   |
| Protein             | Community     | 1.0                    | 0.7 (0.4–1.3) | 1.2 (0.7–2.0) | 0.8 (0.5–1.4) | 0.6 (0.4–1.1) | 0.212                             |
|                     | Biopsy        | 1.0                    | 0.7 (0.4–1.2) | 0.8 (0.5–1.3) | 0.7 (0.4–1.3) | 0.9 (0.5–1.6) | 0.584                             |
| Total fat           | Community     | 1.0                    | 1.0 (0.6–1.7) | 2.0 (1.2–3.5) | 1.5 (0.9–2.5) | 1.5 (0.9–2.5) | 0.072                             |
|                     | Biopsy        | 1.0                    | 0.5 (0.2–0.9) | 1.0 (0.5–1.9) | 0.6 (0.4–1.2) | 1.1 (0.6–2.1) | 0.371                             |
| Saturated fat       | Community     | 1.0                    | 1.2 (0.7–2.1) | 1.0 (0.6–1.7) | 1.2 (0.7–1.9) | 1.3 (0.8–2.3) | 0.331                             |
|                     | Biopsy        | 1.0                    | 1.1 (0.6–2.1) | 0.7 (0.4–1.2) | 1.0 (0.5–1.8) | 1.0 (0.6–1.8) | 0.752                             |
| Monounsaturated fat | Community     | 1.0                    | 1.4 (0.8–2.4) | 1.2 (0.7–2.1) | 1.5 (0.9–2.7) | 1.5 (0.9–2.5) | 0.139                             |
|                     | Biopsy        | 1.0                    | 0.7 (0.4–1.3) | 0.5 (0.3–0.9) | 0.8 (0.4–1.4) | 0.7 (0.4–1.3) | 0.480                             |
| Polyunsaturated fat | Community     | 1.0                    | 0.9 (0.5–1.5) | 1.1 (0.7–1.9) | 1.0 (0.6–1.6) | 1.1 (0.7–1.9) | 0.528                             |
|                     | Biopsy        | 1.0                    | 0.7 (0.4–1.3) | 0.9 (0.5–1.5) | 0.8 (0.4–1.4) | 1.6 (0.9–2.9) | 0.147                             |
| P/S ratio           | Community     | 1.0                    | 0.7 (0.4–1.2) | 0.7 (0.4–1.3) | 0.7 (0.4–1.3) | 0.9 (0.6–1.5) | 0.769                             |
|                     | Biopsy        | 1.0                    | 1.0 (0.6–1.8) | 1.4 (0.7–2.6) | 1.0 (0.6–1.8) | 1.6 (0.9–2.8) | 0.147                             |
| Cholesterol         | Community     | 1.0                    | 1.0 (0.6–1.8) | 1.0 (0.6–1.6) | 1.0 (0.6–1.7) | 0.5 (0.3–0.9) | 0.036                             |
|                     | Biopsy        | 1.0                    | 1.5 (0.9–2.8) | 0.9 (0.5–1.6) | 0.8 (0.5–1.4) | 0.7 (0.4–1.3) | 0.061                             |

<sup>a</sup> Also adjusted for variables shown in footnote to Table 1.<sup>b</sup> Reference category.Table 3 Adjusted relative risks<sup>a</sup> and 95% confidence intervals for the associations between daily retinol and beta-carotene intake and benign proliferative epithelial disorders of the breast

| Dietary item | Control group | RR (95% CI) for level: |               |               |               |               | P value for linear trend (1 d.f.) |
|--------------|---------------|------------------------|---------------|---------------|---------------|---------------|-----------------------------------|
|              |               | 1 (Low) <sup>b</sup>   | 2             | 3             | 4             | 5 (High)      |                                   |
| Retinol      | Community     | 1.0                    | 1.2 (0.7–2.2) | 0.8 (0.5–1.4) | 0.6 (0.3–1.1) | 0.5 (0.3–1.0) | 0.002                             |
|              | Biopsy        | 1.0                    | 1.1 (0.6–1.8) | 1.5 (0.8–2.7) | 1.1 (0.6–2.0) | 1.1 (0.6–2.0) | 0.655                             |
| β-carotene   | Community     | 1.0                    | 0.5 (0.3–0.9) | 0.6 (0.3–1.0) | 0.5 (0.3–0.8) | 0.5 (0.3–0.8) | 0.016                             |
|              | Biopsy        | 1.0                    | 0.7 (0.4–1.4) | 0.6 (0.3–1.0) | 0.8 (0.5–1.4) | 0.8 (0.4–1.4) | 0.439                             |

<sup>a</sup> Adjusted for variables shown in footnote to Table 1.<sup>b</sup> Reference category. Quintiles for categorization of each item were as follows: retinol (μg/day), 245.8, 363.8, 590.6, 1473.2; β-carotene (μg/day), 3284.7, 4474.1, 6138.9, 8262.8.Table 4 Adjusted relative risks<sup>a</sup> and 95% confidence intervals for the associations between daily sugar, starch and fiber intake and benign proliferative epithelial disorders of the breast

| Dietary item | Control group | RR (95% CI) for level: |               |               |               |               | P value for linear trend (1 d.f.) |
|--------------|---------------|------------------------|---------------|---------------|---------------|---------------|-----------------------------------|
|              |               | 1 (Low) <sup>b</sup>   | 2             | 3             | 4             | 5 (High)      |                                   |
| Sugar        | Community     | 1.0                    | 0.8 (0.5–1.4) | 0.8 (0.5–1.4) | 0.5 (0.3–0.9) | 0.6 (0.3–1.0) | 0.010                             |
|              | Biopsy        | 1.0                    | 1.4 (0.8–2.5) | 1.2 (0.7–2.0) | 1.3 (0.7–2.2) | 1.5 (0.8–2.6) | 0.317                             |
| Starch       | Community     | 1.0                    | 0.7 (0.4–1.3) | 0.7 (0.4–1.2) | 0.8 (0.5–1.3) | 0.7 (0.4–1.1) | 0.204                             |
|              | Biopsy        | 1.0                    | 1.9 (1.1–3.4) | 1.6 (0.9–2.7) | 2.3 (1.3–4.1) | 1.7 (1.0–2.9) | 0.024                             |
| Fiber        | Community     | 1.0                    | 1.0 (0.6–1.7) | 0.7 (0.4–1.2) | 0.6 (0.3–1.1) | 0.7 (0.4–1.1) | 0.041                             |
|              | Biopsy        | 1.0                    | 1.2 (0.7–2.2) | 1.2 (0.6–2.2) | 0.6 (0.3–1.0) | 1.4 (0.8–2.6) | 0.752                             |

<sup>a</sup> Adjusted for variables shown in footnote to Table 1.<sup>b</sup> Reference category. Quintiles for categorization of each item were as follows: sugar (g/day), 79.4, 106.4, 132.8, 166.3; starch (g/day), 67.2, 82.5, 100.7, 121.4; fiber (g/day), 13.4, 17.7, 21.8, 26.8.

With biopsy controls as the comparison group, all point estimates of effect (above baseline risk) for  $\beta$ -carotene were less than unity, but the trend in risk was not statistically significant. Additional control for energy intake abolished the statistical significance of the trend in risk for  $\beta$ -carotene ( $P = 0.141$ ) but changed the point estimates for this and the other variables relatively little.

**Carbohydrates.** Risk of BPED in association with intake of sugar (mono- and disaccharides), starch (polysaccharides), and fiber is shown in Table 4. With community controls as the comparison group, risk of BPED was reduced in association with relatively high levels of intake of these factors, and the trends in risk for sugar and fiber were statistically significant; after adjustment for energy intake, however, these trends were no longer statistically significant. When cases were compared with biopsy controls, the variation in risk across levels of intake of the carbohydrates was irregular.

Nevertheless, risk of BPED was increased at the upper levels of starch intake, and the test for linear trend was statistically significant. This effect disappeared after adjustment for energy intake (for trend,  $P = 0.273$ ), while risk of BPED decreased with increasing energy-adjusted fiber intake (point estimates of effect and 95% CI at the 2nd, 3rd, 4th, and 5th quintile levels were, respectively, 0.7 (0.4–1.3), 0.6 (0.3–1.1), 0.6 (0.3–1.2), and 0.6 (0.3–1.2) (for trend,  $P = 0.157$ ).

**Diet-related Risk of BPED by Degree of Atypia.** With community controls as the comparison group, risk of BPED showing a mild degree of atypia decreased with increasing intake of energy, total fat, retinol, and  $\beta$ -carotene (Table 5), while for BPED showing a severe degree of atypia, the variation in risk across levels of these factors was generally not dose-dependent. Adjustment for energy intake removed the trends in risk for retinol and  $\beta$ -carotene in association with mild atypia. When cases were compared with biopsy controls, risk of BPED with a mild degree of atypia increased with increasing energy and fat intake and decreased with increasing  $\beta$ -carotene intake, but the trends were not statistically significant. At severe degrees of atypia, risk was increased at the uppermost level of energy and fat intake, but the latter trend disappeared after adjustment for energy intake.

## DISCUSSION

Studies of biopsy-confirmed BBD (and, indeed, studies of other putative premalignant conditions) are prone to selection

bias, since an unknown proportion of women with BBD comes to clinical attention, and of those women who do, an unknown proportion proceeds to biopsy. In the present study, two control groups were selected in an attempt to address the problem of these biases. Community controls were matched to cases on age and socioeconomic grading of area of residence, the latter matching criterion being an attempt to provide a comparison group from the source population for the cases (*i.e.*, a group of women who, had they developed BBD and been biopsied, would have had their biopsies sent to the same laboratory as the cases). Biopsy controls made up an unmatched group ascertained from the same source as the cases (namely, the pathology laboratory) and, therefore, had been subjected to the same selection processes as the cases. As such, they represent the only potential source of "control" for any selection or referral bias that might be operating here and also provide some control for recall bias. Nevertheless, it is possible that neither of these control groups was appropriate: the community controls may have formed an inappropriate comparison group since the case series may not have been representative of all cases arising in Adelaide during the course of the study (19), while use of the biopsy control group may have resulted in "overmatching" (20). Furthermore, it is possible that overmatching also resulted from use of the community controls, since socioeconomic grading of area of residence may be a crude proxy for dietary habits. However, selection of a random sample of community controls, without matching on socioeconomic status, would not have been appropriate since, as indicated earlier, the case series represented an unknown proportion of all cases. Indeed, in a study of this kind, a true population-based series of cases is not attainable even if all of the pathology laboratories in a given region participate, since symptom-response behavior by individual women varies as a function of social class, age, and so on. In general, in studying BBD and other conditions in which the possibility of serious underdiagnosis exists, use of more than one control group seems justified, because it has been shown that in the absence of estimates of exposure-specific detection rates and exposure-specific rates of special treatment (*e.g.*, biopsy), it is not possible to determine *a priori* which control group will give the correct results (21). The basic principle, however, is to select controls from the same putative cohort as the cases so that, should a control develop the disease of interest, she would be present in the case series.

Table 5 Adjusted relative risks<sup>a</sup> and 95% confidence intervals for the associations between daily energy, total fat, retinol, and  $\beta$ -carotene intake and benign proliferative epithelial disorders of the breast

| Dietary item      | Control group | Degree of atypia <sup>b</sup> | RR (95% CI) for level: |               |               | P value for linear trend (1 d.f.) |
|-------------------|---------------|-------------------------------|------------------------|---------------|---------------|-----------------------------------|
|                   |               |                               | 1 (Low) <sup>c</sup>   | 2             | 3 (High)      |                                   |
| Energy            | Community     | Mild                          | 1.0                    | 0.7 (0.4–1.2) | 0.4 (0.2–0.7) | <0.001                            |
|                   |               | Severe                        | 1.0                    | 0.9 (0.4–2.0) | 1.0 (0.5–2.3) | 0.986                             |
|                   | Biopsy        | Mild                          | 1.0                    | 1.2 (0.7–1.8) | 1.4 (0.9–2.3) | 0.165                             |
|                   |               | Severe                        | 1.0                    | 1.0 (0.5–1.9) | 1.8 (1.0–3.3) | 0.083                             |
| Total fat         | Community     | Mild                          | 1.0                    | 0.8 (0.5–1.3) | 0.5 (0.3–0.9) | 0.015                             |
|                   |               | Severe                        | 1.0                    | 0.8 (0.3–1.8) | 1.0 (0.4–2.3) | 0.939                             |
|                   | Biopsy        | Mild                          | 1.0                    | 1.5 (0.9–2.4) | 1.5 (0.9–2.5) | 0.081                             |
|                   |               | Severe                        | 1.0                    | 0.9 (0.5–1.7) | 2.1 (1.1–3.9) | 0.027                             |
| Retinol           | Community     | Mild                          | 1.0                    | 0.8 (0.5–1.2) | 0.6 (0.4–1.0) | 0.071                             |
|                   |               | Severe                        | 1.0                    | 1.8 (0.7–4.7) | 0.6 (0.3–1.5) | 0.220                             |
|                   | Biopsy        | Mild                          | 1.0                    | 1.3 (0.8–2.1) | 1.2 (0.7–1.9) | 0.475                             |
|                   |               | Severe                        | 1.0                    | 1.6 (0.9–3.0) | 1.4 (0.7–2.6) | 0.284                             |
| $\beta$ -carotene | Community     | Mild                          | 1.0                    | 0.5 (0.3–0.9) | 0.5 (0.3–0.9) | 0.020                             |
|                   |               | Severe                        | 1.0                    | 1.3 (0.6–2.8) | 0.7 (0.3–1.4) | 0.259                             |
|                   | Biopsy        | Mild                          | 1.0                    | 0.6 (0.4–1.0) | 0.8 (0.5–1.3) | 0.279                             |
|                   |               | Severe                        | 1.0                    | 0.8 (0.4–1.5) | 1.2 (0.6–2.3) | 0.647                             |

<sup>a</sup> Adjusted for variables shown in footnote to Table 1.

<sup>b</sup> Mild atypia = grades 0–3 (see "Methods"); severe atypia = grades 4–6.

<sup>c</sup> Reference category. Tertiles for categorization of each item were as follows: energy (kJ/day), 7246.2, 9447.6; total fat (g/day), 70.3, 99.5; retinol ( $\mu$ g/day), 319.5, 1175.2;  $\beta$ -carotene ( $\mu$ g/day), 4051.9, 6829.6.

Additional selection bias may have been introduced into the study as a result of nonresponse by cases and controls. For bias to have arisen from this source, it would be necessary to postulate that the proportion (of cases and/or of controls) "exposed" among nonparticipants differed from the proportion exposed among participants. There are no *a priori* grounds for suspecting that exposure and participation were related, but this possibility cannot be excluded.

Misclassification of study subjects with respect to exposure and/or disease status could also have influenced the results. With respect to disease status, it is possible that some community controls (and, indeed, some biopsy controls) had undetected BPED as a result of which estimates of effect might have been biased conservatively. However, when this issue was addressed (to a limited extent) by excluding women with a history of unbiopsied breast lump, the results were largely unchanged. One other clue to the likely extent of undetected BPED in the community controls comes from the results of autopsy studies. For example, in a recent autopsy study in women of comparable age to those included in the present study, atypical ductal hyperplasia was present in 8% of cases (22), while in an autopsy study in women aged 13–88 years (23), cystic disease with epithelial proliferation was present in about 14% of cases. If comparable figures apply in the present study population, they suggest that misclassification with respect to disease status in the community controls is unlikely to have been a major source of what is, anyway, conservative bias. With respect to the measurement of dietary factors, although there is evidence to support both the repeatability (13, 14) and the criterion validity (10, 14) of measurements of dietary intake obtained by use of food frequency questionnaires, it should be assumed that their intrinsic limitations lead to some individuals being misclassified with respect to intake. In the present study, however, several steps were taken to increase the likelihood that any such misclassification was distributed nondifferentially between cases and controls, as a result of which the estimates of effect would, again, have been biased conservatively (24). First, to minimize the opportunity for dietary change subsequent to disease diagnosis [as recall of past habits may be influenced by current dietary practices (25)], cases were interviewed as soon as possible after diagnosis. Second, cases and controls were instructed to ignore recent changes in dietary habits and to record those which had been their usual practice. Third, the majority of the cases and their matched controls were interviewed at the same time of the year (*i.e.*, in the same season), to minimize the possibility that seasonal variability in food availability may have differentially influenced current dietary practices and hence recall of past practices; also, when estimates of effect derived from comparisons of cases and biopsy controls were adjusted for season of interview, they differed little from those presented here.

Insufficient heterogeneity of exposure in a study population can impair the ability to detect differences between groups. This explanation does not seem likely to account for the results in the present study, in which there was a considerable range of dietary habits. For example, in the community controls, the proportion of calories from fat ranged from about 30% at the 10th percentile of the distribution of this variable to about 46% at the 90th percentile.

The results of animal experiments (26, 27) and *in vitro* studies (28, 29) suggest that dietary fat stimulates the proliferation of mammary epithelium. In support of these observations, fatty foods (*i.e.*, foods containing at least 10% fat) were associated with increased risk of atypical forms of BBD (*i.e.*, those BBD

showing mild to severe proliferation and atypical changes) in a recent case-control study in Israel (30). In contrast, however, the results of the present study do not suggest a role for dietary fat (and energy and protein) in the etiology of BPED. Macronutrient (including carbohydrate) intake by the cases was intermediate between that of the community controls (whose intake was higher, albeit only slightly) and the biopsy controls, and the differences that were observed seem largely attributable to differences in energy intake.

Other than chance and bias, there are at least two possible interpretations of the patterns observed for energy and fat intake. One possibility is that a relatively low intake of energy (or fat) increases risk of nonproliferative BBD. The other possibility (perhaps of more interest) is that BPED and nonproliferative BBD (and also breast cancer) are manifestations of the same syndrome for which there is a specific (but, as yet, unidentified) "proneness." We hypothesize, on the basis of these data, that among women prone to the syndrome, those whose energy or fat intake is close to the population mean will develop proliferative disease, while energy or fat intake at lower levels will decrease the likelihood of BPED and increase the likelihood of nonproliferative forms of BBD, possibly a more benign manifestation of the disease appearing at lower levels of exposure. Also, above average intake will be associated with increased risk of breast cancer. If proneness is genetically determined, then, given the positive association between BBD and breast cancer (31), evidence supporting this hypothesis might come from examination of the dietary patterns and distribution of BBD by histological type in families at high risk of breast cancer.

The present study has provided some evidence to suggest that retinol,  $\beta$ -carotene, and fiber intake are related inversely to risk of BPED, raising the possibility that any influence of these three dietary factors on risk of breast cancer is achieved by a reduction in the frequency of occurrence of BPED. This seems plausible for the retinoids, which appear to be involved in the control of cell differentiation and proliferation and for which there is animal experimental, *in vitro*, and human evidence to suggest a preventive role in mammary carcinogenesis (32, 33). The mechanism by which  $\beta$ -carotene might exert an antineoplastic effect is less clear (34). Nevertheless,  $\beta$ -carotene was inversely associated with risk of breast cancer in a case-control study conducted contemporaneously with the one reported here (35). In the same case-control study of breast cancer (35), there was also some suggestion of an inverse association with fiber intake. Individuals with a high fiber intake have been observed to have relatively low plasma levels of estrogen (1), a consequence of which might be an antiproliferative effect on mammary epithelium.

In conclusion, the results of this study do not suggest that dietary energy and fat intake are associated with increased risk of BPED but provide qualified support for protective effects of  $\beta$ -carotene, retinol, and fiber. As such, they furnish additional evidence against roles for dietary energy and fat intake in breast cancer etiology, while suggesting a route by means of which vitamin A and fiber intake might reduce risk of breast cancer. However, the methodological difficulties described earlier suggest the need for further investigation under circumstances that would allow a reduction or avoidance of the selection and misclassification biases to which studies of conditions such as BPED of the breast are prone. In this regard, case-control studies within populations screened for breast changes (so that controls can be selected from among those found, on screening, not to have evidence of BBD) might prove useful. Indeed, we

are planning to undertake such a study within the National Breast Screening Study (36).

## ACKNOWLEDGMENTS

We thank Dr. N. G. Carter for permission to use the laboratory facilities at Gribbles; R. Pearce for administrative help; S. G. Stevens for the histological preparations; S. Record for assistance with computing; and C. Powell, E. Raymond, P. Rogers, and M. Juras for assistance with interviewing of the study subjects.

## REFERENCES

1. Rohan, T. E., and Bain, C. J. Diet in the etiology of breast cancer. *Epidemiol. Rev.*, 9: 120-145, 1987.
2. Ernster, V. L. The epidemiology of benign breast disease. *Epidemiol. Rev.*, 3: 184-202, 1981.
3. Wang, D. Y., and Fentiman, I. S. Epidemiology and endocrinology of benign breast disease. *Breast Cancer Res. Treat.*, 6: 5-36, 1985.
4. Rohan, T. E., and Cook, M. G. Alcohol consumption and risk of benign proliferative epithelial disorders of the breast in women. *Int. J. Cancer*, 43: 631-636, 1989.
5. Australian Electoral Office. A Quantitative Assessment of Electoral Enrolment in Australia. Research Report. Canberra: Australian Electoral Office, 1983.
6. Stimson, R. J., and Cleland, E. A. A Socio-Economic Atlas of Adelaide. An Analysis of the 1971 Census. Adelaide: School of Social Sciences (Flinders University of South Australia) and Department of Community Welfare, 1975.
7. Bonser, G. M., Dossett, J. A., and Jull, J. W. Human and Experimental Breast Cancer. London: Pitman Medical, 1961.
8. Page, D. L., Zwag, R. V., Rogers, L. W., *et al.* Relation between component parts of fibrocystic disease complex and breast cancer. *J. Natl. Cancer Inst.*, 61: 1055-1063, 1978.
9. Azzopardi, J. G. Problems in breast pathology. In: J. L. Bennington (ed.), *Major Problems in Pathology*, Vol II. London: W. B. Saunders, 1979.
10. Baghurst, K. I., and Baghurst, P. A. The measurement of usual dietary intake in individuals and groups. *Trans. Menzies Found.*, 3: 139-160, 1981.
11. Baghurst, K. I., and Record, S. J. Intakes and sources, in selected Australian populations, of dietary constituents implicated in the etiology of chronic diseases. *J. Food Nutr.*, 40: 1-15, 1983.
12. Thomas, S., and Corden, M. Tables of Composition of Australian Foods. Canberra: Australian Government Printing Service, 1970.
13. Rohan, T. E., Record, S. J., and Cook M. G. Repeatability of estimates of nutrient and energy intake: the quantitative food frequency approach. *Nutr. Res.*, 7: 125-137, 1987.
14. Willett, W. C., Sampson, L., Stampfer, M. J., *et al.* Reproducibility and validity of a semiquantitative food frequency questionnaire. *Am. J. Epidemiol.*, 122: 51-65, 1985.
15. Baghurst, K. I., and Record, S. J. A computerised dietary analysis system for use with diet diaries or food frequency questionnaires. *Community Health Stud.*, 8: 11-18, 1984.
16. Paul, A. A., and Southgate, D. A. T. McCance and Widdowson's The Composition of Foods, Ed. 4. London: Her Majesty's Stationary Office, 1978.
17. Breslow, N. E., and Day, N. E. Statistical Methods in Cancer Research, Vol I: The Analysis of Case-Control Studies, Lyon, France, IARC Scientific Publications, 1980.
18. Willett, W., and Stampfer, M. J. Total energy intake: implications for epidemiologic analyses. *Am. J. Epidemiol.*, 124: 17-27, 1986.
19. Berkowitz, G. S., Canny, P. F., Vivolsi, V. A., *et al.* Cigarette smoking and benign breast disease. *J. Epidemiol. Community Health*, 39: 308-313, 1985.
20. Vessey, M. P. Comment. *J. Chronic Dis.*, 32: 64-66, 1979.
21. Greenland, S., and Neutra, R. An analysis of detection bias and proposed corrections in the study of estrogens and endometrial cancer. *J. Chronic Dis.*, 34: 433-438, 1981.
22. Nielsen, M., Thomsen, J. L., Primdahl, S., *et al.* Breast cancer and atypia among young and middle-aged women: a study of 110 medicolegal autopsies. *Br. J. Cancer*, 56: 814-819, 1987.
23. Frantz, V. K., Pickren, J. W., Melcher, G. W., and Auchincloss, H., Jr. Incidence of chronic cystic disease in so-called normal breasts. *Cancer (Phila.)*, 4: 762-783, 1951.
24. Blettner, M., and Wahrendorf, J. What does an observed relative risk convey about possible misclassification? *Methods Inf. Med.*, 23: 37-40, 1984.
25. Rohan, T. E., and Potter, J. D. Retrospective assessment of dietary intake. *Am. J. Epidemiol.*, 120: 876-887, 1984.
26. Benson, J., Lev, M., and Grand, C. G. Enhancement of mammary fibroadenomas in the female rat by a high fat diet. *Cancer Res.*, 16: 135-137, 1956.
27. Zhang, L., Bird, R. P., and Bruce, W. R. Proliferative activity of murine mammary epithelium as affected by dietary fat and calcium. *Cancer Res.*, 47: 4905-4908, 1987.
28. Kidwell, W. R., Knazek, R. A., Vonderhaar, B. K., and Losonczy, I. Effects of unsaturated fatty acids on the development and proliferation of normal and neoplastic breast epithelium. In: M. S. Arnott, J. van Eys, and Y.-M. Wang (eds.), *Molecular Interrelations of Nutrition and Cancer*, pp. 219-236. New York: Raven Press, 1982.
29. Balakrishnan, A., Cramer, S., Bandyopadhyay, G. K., *et al.* Differential proliferative response to linoleate in cultures of epithelial cells from normal human breast and fibroadenomas. *Cancer Res.*, 49: 857-862, 1989.
30. Lubin, F., Wax, Y., Ron, E., *et al.* Nutritional factors associated with benign breast disease etiology: a case-control study. *Am. J. Clin. Nutr.*, 50: 551-556, 1989.
31. Kelsey, J. L., and Berkowitz, G. S. Breast cancer epidemiology. *Cancer Res.*, 48: 5615-5623, 1988.
32. Sporn, M. B., and Roberts, A. B. Role of retinoids in differentiation and carcinogenesis. *Cancer Res.*, 43: 3034-3040, 1983.
33. Katsouyanni, K., Willett, W., Trichopoulos, D., *et al.* Risk of breast cancer among Greek women in relation to nutrient intake. *Cancer (Phila.)*, 61: 181-185, 1988.
34. Peto, R., Doll, R., Buckley, J. D., *et al.* Can dietary  $\beta$ -carotene materially influence human cancer rates? *Nature (Lond.)*, 29: 201-208, 1981.
35. Rohan, T. E., McMichael, A. J., and Baghurst, P. A. A population-based case-control study of diet and breast cancer in Australia. *Am. J. Epidemiol.*, 128: 478-489, 1988.
36. Miller, A. B., Howe, G. R., and Wall, C. The national study of breast cancer screening: protocol for a Canadian randomized controlled trial of screening for breast cancer in women. *Clin. Invest. Med.*, 4: 227-258, 1981.